Internet's skeletons are awake

You'd think the internet is online because of a dense web of cables plugged into routers and switches. I'd argue it's software, and the unshakable optimism of people maintaining the codebase who thought "we'll fix it later" as its maintenance strategy over the decades.

Well, "later" has finally arrived. It has brought AI agents with an extraordinary talent for finding every piece of bug that has ever existed in the internet. October delivered a spectacular demonstration of just that.

Researchers using frontier AI helped expose two critical vulnerabilities in PHP, the software language used by nearly 7 in 10 websites online right now. These weren't shiny new mistakes. Their roots stretched back years. Suddenly, AI turned weaknesses that had survived generations of developers into working exploits at astonishing speed. Emergency patches followed, but that didn't stop several international government websites from going dark.

And PHP was merely the opening act. Anthropic says its vulnerability-hunting effort had disclosed 6,157 flaws across 591 open-source projects by early October. Its Mythos research showed that frontier models could uncover and exploit weaknesses in major browsers and operating systems. Google's threat researchers documented criminals assembling an AI-assisted credential-harvesting operation in under six hours.

Then came the truly uncomfortable stories. Britain's AI Security Institute reported agents, tested with safety filters disabled, taking unauthorised actions against real internet targets. One tried slipping malicious code into an open-source project. It's a scary development.

Anthropic separately disclosed experimental agents reaching real third-party systems because test environments were misconfigured. India's CERT-In warned that agents had accessed confidential records in Indian organisations, with victims failing to notice. Meanwhile, researchers traced an AI-assisted criminal campaign against more than 400 Latin American banks and financial institutions. American banks, too, reported unvetted third-party agents turning up inside customer accounts.



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Jayesh Shinde
Executive Editor

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But it's not quite an apocalypse, tweeted Nikesh Arora, CEO of Palo Alto Networks in late October. It's an engineer's weary inventory of everything executives have postponed, he said. Finding vulnerabilities is becoming cheap, but fixing them is not.

Banks still depend on COBOL-driven mainframes. Electricity networks use SCADA systems and programmable controllers. Airlines operate decades-old reservation platforms. Hospitals have connected medical equipment that cannot casually be upgraded without potentially interrupting care. All of it talks to modern cloud services, mobile apps, APIs and whatever passes off as SaaS these days.

The various enterprises and software systems that power our modern digital civilization over the internet isn't one codebase awaiting one heroic patch. It's an elaborate Jenga tower of dependencies, customisations and undocumented decisions. Pull the wrong block and disaster strikes.

In this reality, Arora suggests direct patching vulnerabilities wherever possible, while virtually patching to block attacks while upgrades catch up, and the need for a proper integrated security operation. He also punctures the idea that swarms of defensive agents will magically repel offensive swarms. An attacker needs one narrow opening and a few seconds to cause maximum damage. Defenders don't have that luxury.

My worry isn't that AI will wake up and decide to destroy the internet. It's that AI has already awakened something more mundane, the compound interest on decades of delayed technical debt. We can teach machines to discover vulnerabilities really fast. But can we ask organisations to take decisions at anything faster than committee speed? Because on the line is our very means of staying connected, and there's no other alternative in sight.

Let me know your thoughts on this column at: **jayesh.shinde@digit.in** | **@jshinde**